

DIAL PLANNING DOCUMENT

# DIAL Groceries & Liquor — Branch Plan

Agency / marketplace grocery & liquor vertical

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**Date:** 2026-08-14

**Status:** Planning — not SoR over v4

**Source:** DIAL\_Groceries\_Liquor\_Branch\_Plan.md

*Formats the existing branch plan only. Authority remains DIAL Consolidated Plan v4 and the Development Agent Pack.*

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# Dial Groceries & Liquor — Branch Integration Plan (Peer-Depth)

**Status:** Plan-phase artifact (**D-52 / D-56**). **Not product SoR.**  
**Date:** 2026-08-14 (enhanced revision)  
**Authority conflict order:** (1) DIAL\_Consolidated\_Plan\_v4.md → (2) DIAL\_Development\_Agent\_Pack.md → (3) Blueprint / WA / promotions companions → (4) this plan. **On any conflict, v4 + D-log win.**  
**Grill:** Founder grill Part 15 embeds dial-grill-locks questions. Run grill before scaffolding money / WA / maps / AI / catalogue / Intelligence for this branch.  
**Repo facts used:**  
packages/{catalogue,payments,delivery,tax,ledger,promotions,ai,queues,search-indexer,shared}, adapters/{psp,fdms,maps,whatsapp}, apps/{gateway-web,worker-temporal,worker-queues}, Pack \$8 Meili shape, tracer matrices E1–E4, DIAL\_WhatsApp\_Flows\_and\_Templates.md, DIAL\_Promotions\_Package\_Design.md patterns.

## Part 0 — How to read this, confidence & research gaps

### 0.1 Purpose and depth bar

This document is the **groceries & liquor vertical companion** to v4. It aims for **peer depth** with DIAL\_Consolidated\_Plan\_v4.md in *section kinds*: market evidence, problem register, commercial/unit economics, tax/compliance with confidence tags, architecture, companion contracts, D-log-style decisions, launch checklist, open questions — adapted to a third Shop vertical under the same ERP. It does **not** reopen locks. It does **not** replace Spare/Tech product SoR. It proposes **G-log draft decisions** pending founder (Part 13) — nothing here silently locks.

### 0.2 Confidence tags (same grammar as v4 §7)

| Tag             | Meaning                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| [Verified]      | Primary instrument, official notice, or locked DIAL D-log / Pack contract already in-repo                         |
| [Reported]      | Credible secondary (law firm, ministry page summary, trade press) — not cross-checked to primary in this revision |
| [Unverified]    | Research lead or common practice claim — <b>treat as false until counsel/ops confirm</b>                          |
| [Founder-input] | Placeholder cell — commercial number or policy the founder must supply                                            |
| [Engineering]   | DIAL implementation posture that does not invent statute                                                          |

**Honest status:** Liquor licensing, remote-sale hours, platform-vs-merchant licence holder, food-safety delivery duties, and CPA perishables carve-outs for electronic cancellation are **predominantly [Unverified] or [Reported]** in this plan. Engineering implements **configurable gates + audit trails**; counsel supplies numbers, copy, and who must hold what licence before customer-open.

### 0.3 What this document adds (relative to the short prior draft)

1. Market & problem register with competitor evidence and agency rationale.

2. Dual-surface product inventories (web + native) + Shop hub + age-gate + catch-weight deferred.
3. Commercial architecture with take-rate ladders, example `amountMinor` journals, spoilage scenarios, IMTT opex, multivendor split.
4. Full Pack-§8-style `grocery_offers_v1` Meili schema + perishables heartbeat + optional Open Food Facts.
5. Money path: E1a reuse, grocery OfferSnapshot fields, PspAdapter, COD USD, escrow decision table.
6. Tax / FDMS / liquor compliance brief (v4 §7 style) + counsel checklist.
7. Delivery / Temporal: slots, cold-chain, age-check-at-door, multi-stop, thawed-goods SOP.
8. WhatsApp companion chapter ( `FLOW_GROCERY_*` ).
9. AI capability contracts + Promptfoo + MetricContract.
10. Security IDOR/RLS/age-gate audit.
11. OSS stitch table (SoR vs pattern).
12. Trains / tracer G0–G6 with filled DoD evidence columns.
13. Proposed G-1...G-n decisions (pending founder).
14. Launch checklist + ops KPIs.
15. Expanded founder grill with answers-or-defaults.

#### 0.4 Reading order for agents

1. v4 Parts 0–1 locks + D-49...D-61
2. Agent Pack §2 non-negotiables + §8 Meili
3. This plan (Parts 1–15)
4. `DIAL_WhatsApp_Flows_and_Templates.md` style for Flow implementation (mirror, do not fork SoR)
5. `dial-grill-locks` → ticket DoD → `dial-tracer-slice` Build

#### 0.5 Research gaps status (honest % vs v4)

| Gap cluster                                                            | Status                                                                                           | Blocks true 100% peer depth     |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|
| Liquor Act primary cite for age / remote delivery / platform liability | <b>[Unverified]</b> pending counsel                                                              | Yes — counsel brief Part 6.10   |
| Bottle-store / retail hours SI application to <i>delivery</i>          | <b>[Reported]</b> Open Council / SI 197 of 2004 summaries — not counsel-confirmed for e-commerce | Yes                             |
| Food safety / cold-chain statutory duties for marketplace agent        | <b>[Unverified]</b>                                                                              | Yes                             |
| CPA § electronic cancellation vs perishables carve-out read-across     | <b>[Reported]</b> via v4 §7.5 framing + Act text leads — counsel must map grocery SKUs           | Yes                             |
| Founder commercial inputs (take rate, delivery fee owner, spoilage %)  | <b>[Founder-input]</b>                                                                           | Yes — unit economics incomplete |
| Escrow threshold policy for grocery                                    | Grill Q8 / G-log                                                                                 | Soft block G1 money design      |

| Gap cluster                             | Status | Blocks true 100% peer depth        |
|-----------------------------------------|--------|------------------------------------|
| Launch geography / informal food policy | Grill  | Soft block catalogue publish rules |

**Self-assessment:** Structural/engineering depth  $\approx$  ~85–90% of v4 section kinds. True **100%** requires founder commercial cells + Zimbabwe counsel opinions on liquor + food + CPA perishables (Part 6.10). Do not claim legal Verified without those opinions.

## 0.6 Hard locks (never reopen)

Agency **D-58** (no principal stock / no `DIAL_OWNED`); Job Reserve / ledger money SoR; **D-49** B2B hide informal; **D-57** USD browse + ZiG-at-checkout; **D-59** FDMS classes; **D-60** IMTT opex + COD USD; official WhatsApp Cloud API only; MapLibre + Nominatim/OSRM/VROOM SoR (**D-44**); AI never writes payable amounts; no Expo customer shell (**C-5**); delivery job SoR = `packages/delivery` + Temporal (**D-45**); promotions = `@dial/promotions` only (**D-42**).

## Part 1 — Market & problem register

### 1.1 Vision sentence (grocery vertical)

**Dial Groceries & Liquor sells certainty that the right licensed merchant's goods arrive in the booked window, at a frozen USD price the customer reviewed, with agency disclosure and door-level age controls for alcohol — without DIAL taking title to inventory.**

Everything in this vertical either produces that certainty (catalogue freshness, slots, cold-chain eligibility, age gates, ledger, FDMS) or distributes it (web, native, WhatsApp). Features that do neither stay off the critical path.

### 1.2 Market evidence [Reported] / [Unverified]

**Framing.** Public Zimbabwe grocery/liquor *marketplace* sizing with auditor-grade TAM is thin. This plan does **not** invent a false-precision TAM. Evidence below is competitive and behavioural — enough to shape product, not enough to replace founder unit-economics inputs.

| Fact / observation                       | Figure / note                                                                                                         | Why it matters                                                                                                | Confidence | Source lead                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------|
| Established grocery delivery brand in ZW | TillPoint (formerly Fresh in a Box) — app claims same-day / "Deliver Now" windows, Harare-centric history since ~2018 | Incumbent <b>principal / first-party</b> fulfilment pattern; DIAL differentiates as <b>multivendor agency</b> | [Reported] | App Store / Play listings (TillPoint by Fresh in a Box) |
| On-demand horizontal delivery            | Tambi marketing: food, groceries, medicine, liquor, packages                                                          | Liquor already appears in competitor messaging — age/hours are table stakes                                   | [Reported] | tambi.mobi marketing                                    |

| Fact / observation         | Figure / note                                                             | Why it matters                                                                                             | Confidence                             | Source lead                                           |
|----------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------|
| Retail marketplace pitch   | Manna Store: connect to Spar / OK / Pick n Pay via riders; EcoCash + card | Multivendor + EcoCash is expected UX in Harare 2026                                                        | [Reported]                             | LinkedIn / company pitch (treat volumes as marketing) |
| Supermarket chains present | OK, Spar, Pick n Pay, local bottle stores                                 | Supply exists; digital aggregation + trust + cash mechanics are the hard parts (same lesson as Spare §1.2) | [Reported]                             | Market common knowledge + competitor copy             |
| Mobile-money checkout      | EcoCash ubiquitous; COD still culturally strong                           | Aligns <b>D-57</b> EcoCash COD buttons + <b>D-60</b> COD USD                                               | [Verified] DIAL lock + market practice | v4 D-57 / D-60                                        |
| WA as commerce surface     | High WA penetration (Spare premise)                                       | Grocery MVP should include WA thin path (grill Q11 default)                                                | [Reported]                             | v4 channel cost \$4.6; Meta Rest of Africa rates      |

**Competitive position (engineering implication).** Incumbents are mostly **owned-inventory** or **single-retailer** apps, or horizontal courier marketplaces. DIAL's wedge is: **one ERP account** already used for Spare/Tech → Shop tab → groceries/liquor with **agency Sold by {Supplier}**, shared Job Reserve / FDMS / delivery SoR, and B2B formal-only filters. Software alone does not win; freshness heartbeat + slot capacity + liquor compliance ops do.

### 1.3 What comparable grocery ventures get wrong (lessons for DIAL)

| Failure mode                                          | Lesson encoded in this plan                                                                       |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Principal inventory dark stores before unit economics | <b>Forbidden</b> without new D-log ( <b>D-58</b> ). Agency only.                                  |
| Stale stock → refunds as primary recovery             | Perishables heartbeat + short <code>stockValidUntil</code> + confirm-required (adapt Spare §2A-1) |
| Ignoring spoilage / failed delivery cost              | Unit economics Part 3 includes spoilage % scenarios                                               |
| Age gate only on UI, not API                          | Server-derived <code>ageGateRequired</code> + audit trail (Part 10)                               |
| Dual money systems for "grocery OMS"                  | Reject Medusa/Saleor/Fleetbase as SoR                                                             |
| Passing IMTT to customers                             | <b>D-60</b> — never a checkout line                                                               |
| Informal food visible to B2B                          | <b>D-49</b> filter at Meili/API                                                                   |

### 1.4 Problem register — grocery/liquor specific

#### 1.4-1. Perishable oversell / stale availability

**Problem:** Fresh/chilled/frozen stock decays faster than Spare TTL defaults; CSV weekly uploads guarantee lies.

**Solution posture:** State not quantity ( `available` / `confirm_required` / `sourcing` ); short TTL (e.g. 4–24h fresh — **[Founder-input]** exact); WA/dashboard **heartbeat** on top movers; shadow failover only for ambient/pantry same-brand where policy allows — **never silent brand/protein swap** without



customer accept.

**Measure:** confirm-failure rate, % orders saved vs refunded, spoilage write-offs charged to whom (grill Q6).

#### 1.4-2. Cold-chain break / thawed goods disputes

**Problem:** Customer claims thawed chicken; courier claims ambient delay.

**Solution:** Job metadata `coldChain`; courier eligibility filter; POD photo + temperature checklist (MVP checklist, not IoT SoR); refund SOP Part 7.6; risk allocation in supplier contract **[Founder-input]**.

#### 1.4-3. Liquor age & hours non-compliance

**Problem:** Platform enables remote sale; enforcement historically targets licensed premises.

**Solution:** Configurable `liquor_order_window`; ATC + checkout + door age-check; `liquor_licence_ref` on supplier; audit trail. Legal who-holds-licence = counsel (**Unverified** until then).

#### 1.4-4. Multivendor cart settlement bugs

**Problem:** One customer payment, N supplier payables + DIAL fee → drift.

**Solution:** Integer `amountMinor` journals; one Intent/Job Reserve; split ledger lines; webhook idempotency; daily reconcile (v4 \$4.2).

#### 1.4-5. Catch-weight price surprise

**Problem:** Sold-by-weight after pick adjusts payable — disputes + AI temptation.

**Solution:** MVP = **packed fixed weights only** (grill Q13 default). Catch-weight = phase 2 with human pricing engine adjust — AI never writes payable.

#### 1.4-6. Informal food safety optics vs D-60 B2C informal

**Problem:** D-60 allows B2C informal Spare visibility; food may need stricter policy.

**Solution:** Product policy flag `informalAllowedCategories`; default recommend formal-first food + formal-only liquor until counsel (grill Q5).

#### 1.4-7. Substitution UX vs agency disclosure

**Problem:** Out-of-stock banana → substitute brand looks like DIAL principal.

**Solution:** Substitution suggestions return **offer ids** only (AI); customer accepts; Sold by remains supplier; price from catalogue engine.

#### 1.4-8. Slot capacity thrash / late delivery CPA risk

**Problem:** CPA delivery timing rights (v4 \$7.5) + unreasonable time tenders.

**Solution:** Capacity-decrement transactional holds; degrade messaging (Pack \$6.22); reschedule Flow; counsel on cancellation fees for perishables.

#### 1.4-9. B2B pantry / fleet grocery informal leak

**Problem:** Same as Spare D-49 but SKU taxonomy differs.

**Solution:** Identical Meili `supplierFormality=formal` filter + `assertB2bMayPurchase` + `leak=0` tests on `grocery_offers_v1`.

#### 1.4-10. Promo cash-out pressure on groceries

**Problem:** High-frequency grocery invites wallet-like credit abuse.

**Solution:** `@dial/promotions` only; promo credit **non-cash-out** (D-42 / \$2B-30); vertical-scoped campaigns.

1.5 Why agency marketplace (not principal grocer)

| Criterion                            | Principal dark store           | Agency marketplace (locked)               |
|--------------------------------------|--------------------------------|-------------------------------------------|
| Title / COGS                         | DIAL inventory                 | Supplier principal (D-58)                 |
| FDMS goods                           | VAT on GMV risk                | GOODS_FORMAL / INFORMAL + DIAL_FEE (D-59) |
| Capital                              | Warehouses, spoilage inventory | Working capital on fees + float policy    |
| Competitive wedge vs TillPoint-class | Race on assortment owned       | Multivendor + ERP trust stack             |
| Reopen cost                          | New D-log + counsel            | N/A — do not scaffold DIAL_OWNED          |

Part 2 — Product dual-surface & scope

2.1 Shop hub placement [Engineering]

Authenticated Dial home (v4 \$1.4): Shop | Services.  
Shop expands:



Prefer routes under apps/gateway-web ( /grocery/\* ) first; split apps/grocery-web only after bundle measurement (Pack ship-less-JS). Brand: Shop tab under Dial + clear branch branding; one account (grill Q12 default).

2.2 In-MVP (launch) — Full Feature MVP

Catalogue & commerce

- Multivendor grocery + liquor; offerSource = MARKETPLACE only.
- Categories (ops taxonomy [Founder-input]): fresh, pantry, beverages, household, liquor (beer / wine / spirits / RTD).
- pricingMode = unit | weight with packed fixed weight only in MVP ( amountMinor per unit or per packed kg step — never float).
- Cold-chain hints: ambient | chilled | frozen | fragile .
- Restricted SKU / ageGateRequired for liquor.
- Multi-vendor cart → one checkout → split settlement.
- Meili grocery\_offers\_v1 ; Postgres catalogue SoR.
- Supplier onboarding formal/informal + liquor licence fields; Catalogue Factory human approve before Meili publish (D-53).
- Promotions via @dial/promotions only.

Checkout & FX (D-57 adapted)

- Browse/cart `displayCurrency = USD`.
- ZiG only at pay step from `fx_daily_rates / fx_rate_id`.
- Required CTAs: **EcoCash** + **COD** buttons (web + WA); other rails per **D-43**.
- COD settle **USD (D-60)**; indicative ZiG on confirm.
- Eighteen-item electronic disclosure + review before pay (v4 \$7.5) — grocery-adapted counsel copy.

### Delivery

- Slot booking + capacity.
- `packages/delivery` + `DeliveryDispatchWorkflow`.
- MapLibre track; OSRM/VROOM.
- Multi-stop when multi-supplier or batching.
- Liquor age-check-at-door activity before POD.

**Surfaces:** web responsive; native Android+iOS (**not Expo**); WA Flows; admin; supplier portal (Mercur vendor-panel **pattern only**).

**Fiscal & money:** Job Reserve / authorize-capture per policy table Part 5; agency FDMS; same outbox drain.

**AI:** drafts only — guided shopping, substitution ids, ranking assist, ops summaries — `MetricContract` + `Promptfoo` + human promote.

## 2.3 Explicitly out of MVP

- DIAL-owned dark store / `DIAL_OWNED`.
- Catch-weight post-pick price adjust (phase 2).
- Subscription meal kits / recipe SoR / third-party grocery OMS as money SoR.
- In-store POS as DIAL principal; click-and-collect unless founder prioritises.
- Alcohol wholesale B2B licensing marketplace (separate counsel track).
- AI-set prices; Simulated Command Centre auto-pay.
- Baileys; Google/Mapbox SoR; Expo customer shell.
- Cross-border grocery; national expansion beyond founder geography.
- Customer wallet as money SoR; physical fiscal printers.

## 2.4 Web screen inventory ( gateway-web )

| Route / screen                  | Purpose              | Notes                                                           |
|---------------------------------|----------------------|-----------------------------------------------------------------|
| <code>/grocery</code>           | Home / categories    | Brand signal + search entry; USD                                |
| <code>/grocery/search</code>    | Meili PLP            | Facets: category, coldChain, ageGate, price, formality (server) |
| <code>/grocery/[offerId]</code> | PDP                  | Sold by {Supplier}; cold-chain badge; age-gate CTA              |
| <code>/grocery/age-gate</code>  | Attestation modal    | Before liquor ATC; audit write                                  |
| <code>/grocery/cart</code>      | Cart                 | Multi-vendor grouping; USD only                                 |
| <code>/grocery/slots</code>     | Slot picker          | Capacity remaining                                              |
| <code>/grocery/checkout</code>  | Review + disclosures | 18-item + agency                                                |

| Route / screen               | Purpose  | Notes                                |
|------------------------------|----------|--------------------------------------|
| /grocery/checkout/pay        | Pay step | EcoCash COD buttons; ZiG for EcoCash |
| /grocery/orders/[id]         | Status   | ERP status                           |
| /grocery/orders/[id]/track   | MapLibre | Read-only Realtime                   |
| /grocery/orders/[id]/returns | Returns  | Perishable / liquor policy           |
| Support deep-link            | Chatwoot | Attach orderId                       |

Responsive desktop + mobile; @dial/design-tokens (Blueprint \$8.0.1).

## 2.5 Native screen inventory (Android + iOS — C-5)

| Screen                        | Notes                                                                      |
|-------------------------------|----------------------------------------------------------------------------|
| Grocery home / categories     | Not Expo shell                                                             |
| Search + facets               | Meili                                                                      |
| PDP                           | Cold-chain + age-gate                                                      |
| Age attestation               | Modal; biometric optional UX only — legal gate is attestation + door check |
| Cart                          | Multi-vendor Sold by groups                                                |
| Slot picker                   |                                                                            |
| Checkout review               | Disclosures                                                                |
| Pay                           | EcoCash   COD + other rails                                                |
| Order status + MapLibre track |                                                                            |
| Support                       | Chatwoot / WA deep-link                                                    |

Courier: **delivery-android (D-44)** — age-check + cold-chain handling UI (not customer Expo).

## 2.6 Liquor age-gate UX policy [Engineering] + counsel numbers [Unverified]

### Engineering chain (mandatory):

1. Browse: liquor offers visible only if feature flag + hours window open (or browse-ok / ATC-blocked — prefer **ATC blocked outside hours**).
2. Add-to-cart: hard gate — DOB attest or “I am 18+” with T&Cs; write `age_attestations`.
3. Checkout: re-confirm; reject if any line `ageGateRequired` and attestation missing/expired.
4. Door: Temporal activity `awaitAgeCheckAtDoor` — courier records pass/fail + method; **prefer no full ID image retention in MVP** unless counsel requires.
5. API: server derives restricted flag from offer — client cannot omit.

**Reported public lead:** minimum purchase age commonly cited as **18** under Liquor Act Chapter 14:12 [Reported — Tripbase / IARD / secondary digests]. **Do not ship “18” as Verified legal copy until counsel cites Act section and remote-delivery application.** Config default may be `min_age=18` as engineering placeholder labelled counsel-pending.

## 2.7 Catch-weight deferred policy

| Mode                               | MVP                                            | Phase 2                                                                           |
|------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------|
| Packed fixed weight (e.g. 1kg bag) | <b>Yes</b> — OfferSnapshot freezes amountMinor | —                                                                                 |
| Catch-weight (scale at pick)       | <b>No</b>                                      | Human pricing-engine adjust + customer notify; new snapshot version; AI forbidden |

## Part 3 — Commercial architecture & unit economics

### 3.1 Pricing engine (deterministic — v4 \$4.1)

Customer line price (USD minor)  
 = supplier net (valid, unexpired)  
 + DIAL margin (category × supplier agreement) [Founder-input ladder]  
 + delivery allocation (band × coldChain class) [Founder-input owner]  
 + payment cost component (method) [modelled, not always shown]  
 ± promotion (@dial/promotions only)

**Never:** AI prose prices; IMTT as customer line (**D-60**); float math.

Agency display: formal VAT-inclusive goods; informal no goods VAT line; Sold by {Supplier}; B2B formal-only (**D-49**).

### 3.2 Take-rate ladder (publish to supply) [Founder-input]

| Band      | GMV / month / supplier (USD) | Commission on goods net                              | Notes                                       |
|-----------|------------------------------|------------------------------------------------------|---------------------------------------------|
| Launch    | —                            | <b>[Founder-input]</b> e.g. target low-teens blended | Start low; earn increases (v4 \$4.4 lesson) |
| Growth    | —                            | <b>[Founder-input]</b>                               | Volume / SLA / heartbeat compliance         |
| Preferred | —                            | <b>[Founder-input]</b>                               | Promoted placement ≠ money SoR change       |

Delivery fee options (pick one in grill Q6):

| Option                                               | Customer sees                    | FDMS                            | Risk                             |
|------------------------------------------------------|----------------------------------|---------------------------------|----------------------------------|
| <b>A. DIAL delivery fee</b><br>(recommended default) | DIAL_FEE line                    | DIAL VAT on fee                 | DIAL ops cost vs courier payable |
| B. Pass-through courier                              | Courier component                | Careful agency characterisation | Looks like logistics principal   |
| C. Supplier-absorbed                                 | "Free delivery" funded in margin | Margin only                     | Hide true cost                   |

### 3.3 Example journals ( amountMinor ) — illustrative only

**Scenario:** Customer cart — Supplier A pantry \$12.00 + Supplier B liquor \$20.00 + DIAL delivery fee \$3.00 = **\$35.00** USD.

amountMinor scale = cents → 3500 total.

**EcoCash capture (after FX to ZiG at pay — ledger still records USD legs + fx\_rate\_id as designed):**

```
// On verified webhook capture (idempotent)
Dr  cash_psp_clearing      3500 USD
Cr  customer_clearing      3500 USD

// Allocate (same txn or follow-on balanced entries)
Dr  customer_clearing      3500
Cr  supplier_payable_A     1200 // goods A – settlement net of commission per contract
Cr  supplier_payable_B     2000
Cr  dial_fee_revenue        300 // delivery fee example
// Commission skim alternative: reduce payables and Cr dial_commission_revenue
```

**IMTT:** book **DIAL opex** expense when PSP invoices / bank applies — **never** Cr customer surcharge line.

**COD:** intent COD → on POD reconcileCodAfterPod → cash\_courier / customer\_clearing per existing delivery patterns; settle USD.

**Failed delivery / spoilage refund (supplier bears — example policy):**

```
Dr  supplier_payable_A    500 // reverse goods
Cr  cash_psp_refund       500 // or customer_clearing credit
```

Exact commission skim vs gross payable = **[Founder-input]** contract template.

### 3.4 Unit economics skeleton (no line may be deleted)

**Per grocery order**

```
+ Customer price (USD minor sum)
- Supplier net cost(s)
- Delivery cost (courier zone + cold-chain premium)
- Failed-attempt cost × failure_rate [Founder-input]
- Spoilage / thaw write-off × spoilage_rate [Founder-input]
- Payment processing (EcoCash / card / COD variance)
- IMTT economic cost on collection + payout legs (opex) [model ~3.5-4% USD round-trip unless counsel/P
- Grocery WHT if counsel says applicable (NOT auto 30% tech D-50)
- Refund/return cost × return_rate (CPA-aware)
- Oversell/failover cost × confirm_fail_rate
- Ops minutes × loaded cost
- AI cost per order (budgeted)
- WA template cost (utility vs marketing – v4 $4.6)
- Guarantee / food-safety provision %
= Contribution per order
```

**Spoilage / failed-delivery scenarios (illustrative sensitivity)**

| Scenario | Spoilage % of chilled GMV        | Failed delivery %              | Contribution impact              |
|----------|----------------------------------|--------------------------------|----------------------------------|
| Base     | <b>[Founder-input]</b> e.g. 1.5% | <b>[Founder-input]</b> e.g. 3% | Model                            |
| Stress   | 4%                               | 8%                             | Often negative at thin take-rate |

| Scenario   | Spoilage % of chilled GMV | Failed delivery %    | Contribution impact |
|------------|---------------------------|----------------------|---------------------|
| Controlled | 0.5% + supplier SLA       | 2% + slot discipline | Target              |

### 3.5 Multivendor split settlement under agency

1. Single customer authorization/capture for order total.
2. Ledger N× `supplier_payable` + `dial_fee_revenue` (+ promo components).
3. Payout via `PspAdapter instructRelease` / batch when escrow; else T+N schedule **[Founder-input]**.
4. Per-supplier FDMS `GOODS_*` + one or more `DIAL_FEE` receipts (**D-59**).
5. Never net informal goods into B2B Valid VAT claims.

### 3.6 Channel costs (reuse v4 §4.6)

Zimbabwe (+263) → Meta “Rest of Africa”: utility ~\$0.0040; marketing ~\$0.0225 per delivered template [Reported — Meta pricing; re-verify]. Prefer session messages; grocery heartbeats as utility inside open windows.

### 3.7 Promotions (D-41a / D-42)

| Campaign      | Grocery use                                      |
|---------------|--------------------------------------------------|
| PLATFORM      | Delivery fee off / first-order credit (non-cash) |
| SUPPLIER_COOP | Supplier-funded % on their SKUs                  |
| REFERRAL      | Optional; fraud graph; no cash-out               |
| FLASH         | Time-boxed; stock heartbeat required             |

## Part 4 — Catalogue / Meili ( `grocery_offers_v1` )

### 4.1 Index choice

**Recommend separate index** `grocery_offers_v1` (not Spare chassis/OEM fields). Unified index with `vertical` filter is acceptable later; separate keeps `searchableAttributes` clean.

Postgres remains catalogue SoR; Meili = customer browse (**Pack §8** doctrine).

## 4.2 Document shape (Pack §8 style)

```
export interface GroceryOfferDocument {
  id: string // offerId
  masterProductId: string
  title: string
  description: string
  brand?: string
  barcode?: string // EAN/UPC when known
  categoryPath: string[] // e.g. ["fresh","dairy"]
  vertical: 'grocery' | 'liquor' // liquor may also be category under grocery – pick one convention
  pricingMode: 'unit' | 'weight'
  priceMinor: number // USD cents per unit OR per packed weight step
  currency: 'USD' // D-57 – no ZiG on browse docs
  packedWeightGrams?: number // MVP packed only
  unitLabel: string // "each" | "500g" | "1L"
  availability: 'available' | 'confirm_required' | 'sourcing'
  coldChain: 'ambient' | 'chilled' | 'frozen' | 'fragile'
  ageGateRequired: boolean
  hasRestrictedSku: boolean
  liquorCategory?: 'beer' | 'wine' | 'spirits' | 'rtd' | 'other'
  deliveryBandId: string
  stockValidUntil: number // unix ms – short for perishables
  offerSource: 'MARKETPLACE' // D-58 – reject DIAL_OWNED
  supplierFormality: 'formal' | 'informal' // D-49
  supplierDisplayName: string // display only – NEVER authorize by client supplierId
  imageUrl?: string
  offEnrichment?: { // optional Open Food Facts – enrichment only
    productName?: string
    nutriscore?: string
    categoriesTags?: string[]
  }
}
// NEVER index raw supplierId for customer search responses
}
```



4.3 Settings

```
{
  "searchableAttributes": [
    "title",
    "description",
    "brand",
    "barcode",
    "categoryPath",
    "supplierDisplayName"
  ],
  "filterableAttributes": [
    "vertical",
    "categoryPath",
    "pricingMode",
    "availability",
    "coldChain",
    "ageGateRequired",
    "hasRestrictedSku",
    "liquorCategory",
    "currency",
    "deliveryBandId",
    "priceMinor",
    "stockValidUntil",
    "offerSource",
    "supplierFormality"
  ],
  "sortableAttributes": ["priceMinor", "stockValidUntil"],
  "displayedAttributes": [
    "id", "masterProductId", "title", "description", "brand", "barcode",
    "categoryPath", "vertical", "pricingMode", "priceMinor", "currency",
    "unitLabel", "packedWeightGrams", "availability", "coldChain",
    "ageGateRequired", "hasRestrictedSku", "liquorCategory", "deliveryBandId",
    "stockValidUntil", "offerSource", "supplierFormality", "supplierDisplayName",
    "imageUrl"
  ]
}
```

**B2B (D-49):** force `supplierFormality = formal` on Meili + offer list APIs + checkout  
`assertB2bMayPurchase`. Regression: `countInformalB2bLeaks(grocery_offers_v1) === 0`.

**Display currency (D-57):** USD only in index and cart APIs.

4.4 Import / heartbeat for perishables

| Path                                        | Use                            | Outcome                                             |
|---------------------------------------------|--------------------------------|-----------------------------------------------------|
| CSV / xlsx supplier import                  | Bulk catalogue                 | Catalogue Factory review queue ( <b>D-53</b> )      |
| Targeted heartbeat (WA utility / dashboard) | Top movers this week           | Refresh <code>stockValidUntil</code> / availability |
| Photo/PDF price list OCR                    | Optional                       | Same <code>SupplierStockRow</code> + source field   |
| Open Food Facts API                         | Optional enrichment by barcode | <b>Not</b> price/availability SoR; ODbL attribution |

**Freshness defaults (engineering proposals — founder/ops override):**

| coldChain      | Default stock TTL | Heartbeat cadence |
|----------------|-------------------|-------------------|
| ambient pantry | 7d                | weekly top-N      |

| coldChain      | Default stock TTL | Heartbeat cadence |
|----------------|-------------------|-------------------|
| chilled        | 24–48h            | daily             |
| frozen         | 48–72h            | daily             |
| liquor ambient | 14d               | weekly            |

On expiry → degrade to `confirm_required` , do not silent-delete (Spare \$2A-1 pattern).

4.5 Restricted / liquor publish gates

Before Meili publish: `offerSource=MARKETPLACE` ; if liquor → `liquor_licence_ref` present & not expired (when policy on); `ageGateRequired=true` ; hours config does not need to hide browse but must block ATC/checkout.

Part 5 — Money path (E1a reuse)

5.1 End-to-end spine

```
Cart (USD lines, amountMinor)
  → assertB2bMayPurchase(formality) per line (D-49)
  → assertAgeGate + liquor hours
  → freeze OfferSnapshot(s) per supplier + DIAL fee + promo components
  → createCheckoutPayment(EcoCash|COD|...) OR authorizeJobReserve (escrow)
  → customer action ≠ truth
  → verified PSP webhook + claimProcessedEvent
  → ledger journal
  → enqueueFiscalReceipt GOODS_* + DIAL_FEE
  → money outbox → FDMS Virtual Gateway
  → createDeliveryJob(s) + slot bind + start DeliveryDispatchWorkflow
```

Reuse: `freezeOfferSnapshot` , `createCheckoutPayment` , `authorizeJobReserve` , `runE1aMoneySpine` , `@dial/tax` — **do not duplicate.**

5.2 Grocery-specific OfferSnapshot fields

| Field                                                   | Purpose                                    |
|---------------------------------------------------------|--------------------------------------------|
| <code>vertical</code>                                   | <code>grocery</code>   <code>liquor</code> |
| <code>supplierDisplayName</code>                        | Agency disclosure freeze                   |
| <code>supplierFormality</code>                          | FDMS class selection                       |
| <code>coldChain</code>                                  | Job metadata                               |
| <code>ageGateRequired</code>                            | Door check + audit link                    |
| <code>slotId</code>                                     | Capacity bind                              |
| <code>packedWeightGrams</code> / <code>unitLabel</code> | Dispute evidence                           |
| <code>pricingMode</code>                                | unit   weight (packed)                     |
| <code>fx_rate_id</code>                                 | When ZiG pay conversion                    |
| <code>promo_campaign_id[]</code>                        | Logged components                          |

| Field                                  | Purpose             |
|----------------------------------------|---------------------|
| liquor_licence_ref (supplier snapshot) | Compliance evidence |

AI **must not** write any payable keys ( `assertNoPayableKeys` ).

### 5.3 PspAdapter rails (D-43)

| Rail                 | Grocery MVP posture                          |
|----------------------|----------------------------------------------|
| EcoCash              | Required button; ZiG payable from daily rate |
| COD                  | Required button; USD settle                  |
| Paynow               | URL / escrow-first ask (D-60)                |
| ContiPay             | As enabled                                   |
| PayPal Orders v2     | As enabled                                   |
| Escrow / Job Reserve | Per decision table below                     |

### 5.4 Escrow vs direct capture — decision table

| Condition                        | Path                                       | Rationale                  |
|----------------------------------|--------------------------------------------|----------------------------|
| COD                              | Direct COD intent → POD reconcile          | D-60                       |
| EcoCash / card below threshold T | Direct capture                             | Speed; low ticket          |
| Multi-vendor OR total ≥ T        | Job Reserve / escrow hold                  | Dispute / split release    |
| Liquor-only high ticket ≥ T_liq  | Escrow preferred                           | Returns / age-fail refunds |
| B2B formal                       | Escrow or Net terms <b>[Founder-input]</b> | TIN capture                |

**T, T\_liq:** grill-accepted **provisional** defaults (ops-tunable config — not statute): **T = 5000** USD minor (\$50.00); **T\_liq = 10000** USD minor (\$100.00). Engineering: strategy interface, not hard-coded magic numbers in adapters.

### 5.5 WHT note (D-50 boundary)

Tech hire 30% ITF263 path stays for **Dial a Tech**. Grocery supplier payouts: **pluggable withholding strategy**; do **not** auto-apply 30% tech WHT; counsel decides marketplace commission/payout WHT (**Unverified** applicability).

## Part 6 — Tax / FDMS / liquor compliance brief

**How to read:** Research, not legal advice. Tags per Part 0.2. Mirror v4 §7 honesty.

### 6.0 Load-bearing inheritance from v4 (do not re-litigate)

| Topic                   | Lock                                                    | Confidence              |
|-------------------------|---------------------------------------------------------|-------------------------|
| Agency characterisation | D-2 / <b>D-58</b>                                       | [Verified] founder lock |
| FDMS receipt classes    | <b>D-59</b> DIAL_FEE / GOODS_FORMAL /<br>GOODS_INFORMAL | [Verified] lock         |

| Topic                                    | Lock        | Confidence                                                        |
|------------------------------------------|-------------|-------------------------------------------------------------------|
| Virtual fiscalisation                    | D-40a       | [Verified] lock                                                   |
| VAT 15.5% from 2026                      | v4 §7.1     | [Verified] in v4 research                                         |
| IMTT = opex                              | <b>D-60</b> | [Verified] lock                                                   |
| COD USD                                  | <b>D-60</b> | [Verified] lock                                                   |
| CPA electronic disclosure + 7-day cancel | v4 §7.5     | [Verified] Act framing in v4 — grocery read-across <b>counsel</b> |
| Job Reserve needs licensed PSP escrow    | C-4 / §7.3  | [Verified] RBZ posture in v4                                      |

## 6.1 VAT & agency FDMS for grocery lines

Same as Spare agency model:

- DIAL taxable supply = **fees/commission** → DIAL\_FEE (+ VAT on fee).
- Formal supplier goods → GOODS\_FORMAL (supplier deemed seller; VAT-inclusive; single invoice on-behalf rules per D-59).
- Informal goods → GOODS\_INFORMAL (no goods VAT line; **never B2B**).
- No GMV VAT split-pot as principal.
- Buyer TIN for B2B Valid claims (v4 §7.1).

## 6.2 Liquor licensing & hours

| Topic                                           | Public research lead                                                                                                            | Tag                                     | Engineering                                                           |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------|
| Governing statute                               | Liquor Act Chapter <b>14:12</b> ; Liquor Licensing under Ministry of Local Government                                           | [Reported]                              | Store licence type + ref + expiry on supplier                         |
| Licence types                                   | Retail, wholesale, restaurant, hotel, bottle store, etc.                                                                        | [Reported] MLG / Act schedule summaries | liquor_licence_type enum ops-configured                               |
| Operating hours                                 | SI summaries e.g. bottle store often cited <b>08:00–20:00</b> ; bars <b>10:00–23:00</b> (Open Council / SI 197 of 2004 digests) | [Reported]                              | liquor_order_window config per band — <b>fail closed</b>              |
| Apply hours to <i>delivery / remote sale</i> ?  | Not established in this research                                                                                                | <b>[Unverified]</b>                     | Assume restricted until counsel says otherwise                        |
| Who must hold licence for marketplace delivery? | Platform vs each merchant                                                                                                       | <b>[Unverified]</b>                     | Default eng: <b>merchant holds</b> ; block liquor publish without ref |
| Sale to minors                                  | Commonly cited 18+; supply to under-18 offence; some digests note guardian-consent nuances                                      | [Reported]                              | Config min_age; door check; <b>counsel copy</b>                       |
| Credit sales of liquor                          | News commentary cites prohibitions on credit liquor sales                                                                       | [Reported]                              | Prefer no “buy now pay later” for liquor SKUs                         |

| Topic                           | Public research lead                          | Tag        | Engineering                     |
|---------------------------------|-----------------------------------------------|------------|---------------------------------|
| National Alcohol Policy reforms | Government review of licensing/hours reported | [Reported] | Keep hours <b>config-driven</b> |

### 6.3 Food safety & delivery notes

| Topic                                                   | Tag                                                                                 | Engineering / ops                                         |
|---------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Mandatory safety/quality standards for goods marketed   | [Reported] CPA s. goods safety themes / v4 product liability                        | Supplier warranties + indemnity; evidence photos          |
| Cold-chain statutory IoT logging                        | <b>[Unverified]</b>                                                                 | MVP = eligibility + checklist; not IoT SoR                |
| Who is “supplier” for unsafe food on agency marketplace | <b>[Unverified]</b> — v4 §7.5 joint-and-several risk applies to platforms in spirit | Contracts + insurance; counsel                            |
| Allergen / labelling                                    | <b>[Unverified]</b> detail                                                          | Display supplier-provided fields; OFF enrichment optional |

### 6.4 Consumer protection — perishables [Reported] / counsel

v4 §7.5: electronic transactions have strong cancellation / disclosure rules [Verified in v4]. Perishable exclusions for some cooling-off contexts are often discussed; **whether they read across to grocery e-commerce cancellation** is a **counsel question** (v4 already flags carve-out ambiguity).

#### Engineering SOP pending counsel:

- Eighteen-item disclosure + review on web/WA/native.
- Clear perishable / opened-liquor return policy in ERP + T&Cs module (§3.8 pattern).
- Thawed-goods refund path (Part 7.6) without waiting for perfect legal taxonomy.
- Do not silently exclude statutory rights in UI copy.

### 6.5 Data protection (age attestations)

Age attestations and optional ID checks are personal data. Apply v4 §7.6 / D-32: minimise retention; prefer boolean + method + timestamp; POTRAZ licence; cross-border AI must not receive ID images / names / phones.

### 6.6 Counsel checklist (grocery/liquor addendum to v4 §7.10)

#### Must confirm before customer-open of liquor SKUs:

1. Minimum age and acceptable ID types for *delivery* handover.
2. Whether Liquor Act hours bind remote acceptance of orders vs only on-premises sale.
3. Whether DIAL as agent needs any liquor licence, or only merchants.
4. Advertising restrictions for alcohol on WA marketing templates.
5. Food business registration / municipal permits for suppliers on platform.
6. CPA cancellation / returns for perishables and alcohol (opened vs unopened).
7. Product liability allocation for unsafe food delivered via DIAL couriers.
8. Whether informal food traders may be B2C-visible (policy + law).
9. WHT / VAT marketplace specifics for grocery commissions (beyond tech D-50).
10. Credit-sale prohibition implications for COD vs postpaid liquor.

6.7 What could not be verified (grocery-specific)

- Primary-section citation for remote liquor delivery age check duties.
- Binding hours for e-commerce liquor order *placement vs handover*.
- Platform liquor licence category (if any).
- Specific municipal food-handler certificate matrix per Harare district.
- Exact CPA perishable carve-out application to § electronic cancellation.
- Competitor GMV / take rates (marketing noise only).

Part 7 — Delivery / Temporal

7.1 Slot capacity model

```
DeliverySlot {
  slotId, bandId, windowStart, windowEnd,
  capacityTotal, capacityReserved, capacityRemaining,
  coldChainAllowed: ambient[] | chilled | frozen,
  liquorAllowed: boolean
}
```

- Checkout binds `slotId` with **transactional** decrement (Postgres); idempotent hold key = `cartId|orderId`.
- Release hold on cancel / payment fail TTL.
- Missed slot → Temporal timer → WA utility notify → reschedule Flow.

7.2 Workflow recommendation

**Extend** `DeliveryDispatchWorkflow` (preferred) — do not invent parallel job SoR.

| Activity                               | When                          |
|----------------------------------------|-------------------------------|
| <code>validateColdChainCapacity</code> | Offer ranking / assign        |
| <code>multiStopPlanVroom</code>        | Multi-supplier / batch        |
| <code>awaitAgeCheckAtDoor</code>       | <code>ageCheckRequired</code> |
| <code>recordHandlingChecklist</code>   | chilled/frozen POD            |
| Existing offer→accept→reassign→FIFO    | Unchanged ( <b>D-45</b> )     |

Algorithm donor: AWS Last Mile Hyperlocal MIT-0 patterns inside `packages/delivery` (**D-45a**) — not Fleetbase.

7.3 Cold-chain eligibility

- Courier profile flags: `canChilled`, `canFrozen`, bag type.
- Job `coldChainMax` = max severity among lines.
- Ineligible couriers filtered before offer.
- MVP: no continuous temperature SoR; optional phase-2 logger.

## 7.4 Age-check-at-door

ageCheckStatus: pending → passed | failed | skipped\_not\_required

- Failed → cancel/return-to-supplier path + refund policy (founder).
- Store: timestamp, courierId, method ( visual\_id | dob\_match | refused ), result — **not** full ID image unless counsel mandates secure upload.

## 7.5 Multi-stop

- Cart with suppliers A+B → either one job multi-stop or N jobs — **[Founder-input]** default: **one job multi-stop when same band/slot**, else split.
- VROOM plan returns stopSequence; customer track shows aggregated ETA honesty.

## 7.6 Refunds for thawed / damaged goods — SOP (ops)

| Step                                                                 | Owner                            | System                                      |
|----------------------------------------------------------------------|----------------------------------|---------------------------------------------|
| 1. Customer reports within policy window                             | Support                          | Ticket + orderId                            |
| 2. Evidence: POD photo + customer photo                              | Support/courier                  | Attachments                                 |
| 3. Classify: thaw_suspected / damaged_transit / wrong_item / quality | Support                          | Taxonomy                                    |
| 4. Liability: courier vs supplier vs DIAL fee absorb                 | Ops rules <b>[Founder-input]</b> | Ledger reason codes                         |
| 5. Refund / replace / partial                                        | Payments                         | Idempotent refund + fiscal credit note path |
| 6. Supplier score / oversell fee if confirm failure                  | Catalogue                        | Statement line                              |

Opened liquor: default **no return** unless defect — counsel copy.

## Part 8 — WhatsApp companion chapter

**Align style with** DIAL\_WhatsApp\_Flows\_and\_Templates.md . Official Cloud API only. Same ERP APIs — WA never second SoR. **D-57**: USD browse/cart; EcoCash|COD **buttons**; **D-58/D-59** fiscal outbox shared.

### 8.1 Happy path

```
Menu → FLOW_GROCERY_HOME
  → FLOW_GROCERY_SEARCH (Meili) — USD prices
  → Age gate screen if liquor add
  → FLOW_GROCERY_CART (USD)
  → FLOW_GROCERY_SLOT
  → FLOW_GROCERY_CHECKOUT (disclosure review)
  → Pay buttons: EcoCash | COD | other
  → Utilities: order_confirmed → packing → out_for_delivery → delivered
  → FLOW_GROCERY_TRACK
```

## 8.2 Flow screen maps

### FLOW\_GROCERY\_HOME

Screens: Welcome · Shop groceries · Liquor (age notice) · Track · Live chat

### FLOW\_GROCERY\_SEARCH

| Screen   | Fields / logic                                                                                                       |
|----------|----------------------------------------------------------------------------------------------------------------------|
| Query    | Text; category chips optional                                                                                        |
| Results  | <code>data_exchange</code> list: title, unitLabel, <b>priceMinor USD</b> , coldChain, Sold by display, ageGate badge |
| PDP      | Detail; ATC                                                                                                          |
| Age gate | If <code>ageGateRequired</code> — attest before ATC write                                                            |

### FLOW\_GROCERY\_CART

Lines grouped by Sold by; edit qty; USD totals; continue / checkout.

### FLOW\_GROCERY\_SLOT

Select window; show remaining capacity; cold-chain notes.

### FLOW\_GROCERY\_CHECKOUT

| Screen    | Purpose                                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Delivery  | Saved place / pin+landmark+phone                                                                                                                |
| Buyer tax | Optional VAT/TIN B2B                                                                                                                            |
| Review    | USD totals, suppliers, cancellation/perishable summary, T&Cs — <b>mandatory</b>                                                                 |
| Pay       | <b>Buttons/CTAs:</b> EcoCash   COD   Paynow/other — <b>not free-text</b> . EcoCash → <code>ZiG + fx_rate_id</code> . COD → USD + indicative ZiG |

### FLOW\_GROCERY\_TRACK

Order id → ERP status (not Chatwoot).

### FLOW\_GROCERY\_RETURNS (MVP)

Reason taxonomy; media; liquor opened block messaging.

### FLOW\_SUPPLIER\_GROCERY\_HEARTBEAT

Supplier Yes/Sold-out on listed SKUs (utility).

## 8.3 `data_exchange` stubs (TypeScript-shaped)

```
type GroceryFlowRequest =
  | { screen: 'GROCERY_SEARCH'; data: { query: string; categoryPath?: string[] } }
  | { screen: 'GROCERY_ADD_CART'; data: { offerId: string; qty: number; ageAttestToken?: string } }
  | { screen: 'GROCERY_SET_SLOT'; data: { cartId: string; slotId: string } }
  | { screen: 'GROCERY_APPLY_PROMO'; data: { code: string; cartId: string } }
  | { screen: 'GROCERY_CHECKOUT_PAY'; data: { orderId: string; method: 'ECOCASH' | 'COD' | 'PAYNOW' |
```

Server re-prices from ERP; never trust client `priceMinor`.



## 8.4 Templates register

| Template name               | Category   | When                     | Body shell                                                   |
|-----------------------------|------------|--------------------------|--------------------------------------------------------------|
| grocery_welcome_menu        | UTILITY    | Opt-in / link            | Hi {{1}}. Groceries, Track, Live chat.                       |
| grocery_order_confirmed     | UTILITY    | Paid                     | Order {{1}} confirmed. Total {{2}} {{3}}. Slot {{4}}.        |
| grocery_supplier_confirming | UTILITY    | Confirm SLA              | Confirming stock for {{1}} by {{2}}.                         |
| grocery_out_for_delivery    | UTILITY    | Courier assigned         | Order {{1}} on the way. Track {{2}}.                         |
| grocery_delivered           | UTILITY    | POD                      | Order {{1}} delivered. Issues? {{2}}                         |
| grocery_age_check_failed    | UTILITY    | Door fail                | Age verification failed for {{1}}. Refund/return per policy. |
| grocery_slot_missed         | UTILITY    | Timer                    | We missed slot {{1}}. Reschedule: {{2}}                      |
| grocery_thaw_followup       | UTILITY    | Cold-chain complaint ack | We received your report on {{1}}. Ticket {{2}}.              |
| grocery_payment_link        | UTILITY    | Resume unpaid            | Pay for {{1}}: {{2}} expires {{3}}.                          |
| grocery_cart_resume         | MARKETING* | Abandoned                | Resume grocery cart {{1}}. STOP to opt out.                  |
| supplier_grocery_heartbeat  | UTILITY    | Stock                    | Still have these SKUs? Yes / Sold out.                       |

\*Prefer free session reminders.

## 8.5 Live chat

Chatwoot handoff with `customer_id`, `cart_id`, `orderId`; status updates **only** via ERP.

## Part 9 — AI ( packages/ai only)

### 9.1 Rules (inherit v4 §5.1 + D-61)

Production = typed capabilities + LiteLLM→Gemini + Temporal/BullMQ. **No prod agent host.** AI never writes payable amounts. D-32 strip identity. Merge gate: `dial-ai-capability-review` (D-56). Promote: Promptfoo + human (D-54).

### 9.2 Capability contracts

| Capability            | Input (Zod)                                | Output                               | Forbidden                                     |
|-----------------------|--------------------------------------------|--------------------------------------|-----------------------------------------------|
| groceryGuidedShopping | dietary prefs, category hints (no raw PII) | structured list draft / category ids | prices, payables                              |
| substitutionSuggest   | unavailable offerId, constraints           | alternate <b>offerIds</b> + reasons  | amounts                                       |
| searchRankingAssist   | query + candidate ids                      | rank hints / synonyms proposal       | silent Meili rewrite without human            |
| opsGrocerySummary     | MetricContract ids + aggregates            | narrative draft                      | attemptCommandCentrePayout ; Simulated≠Actual |

### 9.3 Promptfoo outline

| Golden                 | Assert                                                                            |
|------------------------|-----------------------------------------------------------------------------------|
| Money keys absent      | No <code>price</code> , <code>amountMinor</code> , <code>fee</code> in model JSON |
| Substitution ids exist | <code>ids</code> ∈ catalogue fixture                                              |
| PII egress             | no phone/name/address in request after strip                                      |
| Liquor minor           | no assist that bypasses ageGate                                                   |
| Informal B2B           | no suggest informal offers when <code>buyer_segment=b2b</code>                    |

### 9.4 MetricContract KPIs (examples)

| Metric id                                   | Definition                      | Actual vs Simulated                   |
|---------------------------------------------|---------------------------------|---------------------------------------|
| <code>grocery.fill_rate</code>              | paid orders / checkout starts   | Actual                                |
| <code>grocery.confirm_fail_rate</code>      | supplier confirm fails / orders | Actual                                |
| <code>grocery.slot_ontime_pct</code>        | POD in window / deliveries      | Actual                                |
| <code>grocery.age_check_fail_rate</code>    | door fails / liquor jobs        | Actual                                |
| <code>grocery.spoilage_refund_bps</code>    | thaw refunds / chilled GMV      | Actual                                |
| <code>grocery.contribution_per_order</code> | unit economics                  | Actual; Simulated scenario never pays |

### 9.5 Privacy / egress

Omit identity from outbound payloads; no ID images to Gemini; Presidio on free text; transfer register; unbundled consent for cross-border (v4 \$5.7 / \$7.6).

## Part 10 — Security

### 10.1 IDOR resource list (extend T9)

| Resource                             | AuthZ                                        |
|--------------------------------------|----------------------------------------------|
| <code>grocery_cart</code>            | owner <code>userId</code> from JWT           |
| <code>grocery_order</code>           | buyer or supplier-org scoped                 |
| <code>delivery_slot_hold</code>      | owner                                        |
| <code>age_attestation</code>         | owner; courier write only via job assignment |
| <code>delivery_job</code> / offers   | D-47 patterns                                |
| <code>promo_credit</code>            | owner; no cash-out                           |
| <code>supplier_liquor_licence</code> | supplier org + admin                         |
| Admin catalogue publish              | role + Factory queue                         |

Never trust body `userId` / `email` / `role`. Authorize before cache; cache keys include `userId`.

## 10.2 RLS notes

Supabase RLS by `user_id` / `supplier_org_id` **necessary not sufficient** — object-level `assertResourceAccess` still required (**D-47**).

## 10.3 Age-gate audit trail

```
age_attestations (
  id, user_id, channel, offer_id?, order_id?,
  method, result, min_age_policy_version,
  created_at, ip_hash?, user_agent_hash?
) append-only
```

Courier door check = separate `age_check_events` linked to `delivery_job_id`.

## 10.4 Webhooks & secrets

Signature + idempotency before mutate (PSP, WA, FDMS). No `NEXT_PUBLIC_` / `VITE_` on `service_role`/PSP/WA/FDMS. D-48 Semgrep/Checkov/Renovate; Threat Dragon model for grocery checkout + age-gate; Strix staging only.

## 10.5 B2B informal

Meili + API + checkout; grocery index leak tests.

## Part 11 — OSS stitch table

| Need                | Recommend                                   | License           | SoR vs pattern                                    |
|---------------------|---------------------------------------------|-------------------|---------------------------------------------------|
| Multivendor UX      | Mercur B2C / Medusa marketplace UX          | MIT/BSD           | <b>Pattern only</b> → DIAL APIs                   |
| Supplier panel      | Mercur vendor-panel                         | MIT               | Pattern → supplier-web                            |
| Search              | <b>Meilisearch</b>                          | MIT               | Customer search SoR (Postgres catalogue SoR)      |
| Maps/route/VRP      | MapLibre + Nominatim + OSRM + VROOM         | BSD/GPL self-host | <b>D-44 SoR</b>                                   |
| Delivery UX         | foodhub-compose rider (strip Google/Stripe) | Apache-2.0        | Pattern; job SoR = <code>packages/delivery</code> |
| Dispatch algorithms | AWS Last Mile Hyperlocal                    | MIT-0             | Reimplement in package ( <b>D-45a</b> )           |
| Grocery OMS         | Saleor/Medusa runtime                       | —                 | <b>Rejected</b> as money/job SoR                  |
| Enrichment          | Open Food Facts                             | ODbL              | Enrichment only                                   |
| Images              | Sharp                                       | Apache-2.0        | Tier 1                                            |
| Chat                | Chatwoot                                    | MIT               | Support; status from ERP                          |
| Analytics/flags     | PostHog                                     | MIT               | T8                                                |
| Evals               | Promptfoo + Langfuse                        | MIT/etc.          | D-54                                              |
| Workflows           | Temporal + BullMQ                           | MIT               | Already SoR                                       |
| DB/auth             | Supabase                                    | Apache-2.0        | RLS + Realtime                                    |

| Need                        | Recommend                                          | License | SoR vs pattern                |
|-----------------------------|----------------------------------------------------|---------|-------------------------------|
| Native                      | Kotlin + Swift donors<br>(CoolMallKotlin patterns) | —       | C-5 no Expo customer shell    |
| Promotions                  | Medusa+OfferKit ideas                              | —       | In-repo @dial/promotions only |
| Fiscal                      | In-house FDMS gateway                              | —       | D-40a / D-59                  |
| ZW liquor compliance engine | None suitable                                      | —       | Config + counsel              |

## Part 12 — Trains / tracer G0–G6 (D-52)

**Rule:** Plan (this doc + grill) → Build thin vertical → Expand in-ticket → Done at DoD 100%. Tracer ≠ stub-as-MVP. Fits **T0–T9 / E1–E6** — no new train numbers; no Train 0–10 (**D-53**).

### G0 — Plan residual

| DoD item                     | Acceptance criteria (evidence)                                                        |
|------------------------------|---------------------------------------------------------------------------------------|
| Grill answered               | Part 15 answers recorded in ticket or linked doc                                      |
| Ticket opened                | One thin-vertical ticket with DoD + owner (Dev Manager)                               |
| Counsel liquor brief started | Checklist 6.6 filed or explicitly deferred with flag <b>no liquor SKUs</b> until done |
| Living docs                  | Same PR note when Build lands (Blueprint §8.0.2)                                      |

### G1 — Thin vertical (money + browse)

**Path:** Browse USD → cart → checkout EcoCash|COD → Intent/Job Reserve → webhook → ledger → FiscalReceiptQueued → delivery job create.

| DoD                                                     | Evidence (acceptance)                                                        |
|---------------------------------------------------------|------------------------------------------------------------------------------|
| Meili/stub returns MARKETPLACE grocery offers           | Automated test: query returns docs with <code>offerSource=MARKETPLACE</code> |
| Cart USD only; no ZiG on browse                         | Test asserts <code>currency===USD</code> on cart API                         |
| EcoCash ZiG from daily rate + <code>fx_rate_id</code>   | Test with fixture rate row; snapshot persists <code>fx_rate_id</code>        |
| COD USD + indicative ZiG                                | Test COD confirm payload                                                     |
| OfferSnapshot Sold by {Supplier}; AI cannot set payable | Unit test freeze + <code>assertNoPayableKeys</code> on AI fixtures           |
| B2B informal rejected                                   | Test <code>buyer_segment=b2b</code> + informal offer → 403/empty             |
| Webhook sig + idempotency                               | Replay webhook → single ledger effect                                        |
| FDMS GOODS_* + DIAL_FEE queued                          | Outbox rows asserted by class                                                |
| No DIAL_OWNED path                                      | Reject publish/test for DIAL_OWNED                                           |
| IMTT not on checkout lines                              | Money-path review checklist signed                                           |
| Responsive web checkout                                 | Blueprint §8.0.1 QA note on ticket                                           |

**G2 — Expand commerce**

| DoD                                          | Evidence                                            |
|----------------------------------------------|-----------------------------------------------------|
| Multi-vendor cart + split payables           | Ledger test: 2 suppliers → 2 payables + fee         |
| Weight/unit packed pricing                   | Snapshot includes packedWeightGrams                 |
| Categories + cold-chain badges               | UI + Meili filter tests                             |
| Supplier onboarding + Factory review/publish | Review queue → Meili doc visible only after approve |
| Admin catalogue screens                      | Screen checklist pass                               |

**G3 — Delivery expand**

| DoD                                 | Evidence                                      |
|-------------------------------------|-----------------------------------------------|
| Slots + capacity transactional hold | Concurrent hold test: capacity never negative |
| Cold-chain eligibility filter       | Ineligible courier not offered                |
| Multi-stop VROOM                    | Fixture plan stopSequence length ≥ 2          |
| Customer MapLibre track             | Recon smoke (dial-webapp-recon)               |
| delivery-android handling UI        | Build screenshot / checklist                  |

**G4 — Liquor compliance expand**

| DoD                          | Evidence                                   |
|------------------------------|--------------------------------------------|
| ATC + checkout age gate      | API rejects liquor ATC without attestation |
| Door check blocks POD        | Workflow test: POD forbidden while pending |
| Audit trail rows             | DB assert append-only events               |
| Restricted hours fail closed | Outside window ATC → 403                   |
| Licence publish gate         | Expired licence → cannot publish liquor    |

**G5 — WhatsApp**

| DoD                                     | Evidence                          |
|-----------------------------------------|-----------------------------------|
| FLOW_GROCERY_SEARCH→CART→CHECKOUT→TRACK | Meta draft + ERP integration test |
| EcoCash COD buttons not free-text       | Flow JSON review                  |
| Chatwoot handoff ids                    | Ticket fields present             |
| Same intents as web                     | Shared payment intent id asserted |

**G6 — AI drafts**

| DoD                     | Evidence         |
|-------------------------|------------------|
| Capabilities + Zod      | Typecheck + unit |
| Promptfoo goldens green | CI smoke         |

| DoD                             | Evidence                       |
|---------------------------------|--------------------------------|
| dial-ai-capability-review audit | docs/agent-audits/* attached   |
| Command Centre Simulated banner | UI/assert Simulated cannot pay |

### Completion matrix (AC × channel) — fill Y + evidence link; blank = incomplete

| AC theme                 | Web       | WA              | Native          | Admin/Courier       |
|--------------------------|-----------|-----------------|-----------------|---------------------|
| USD browse/cart          | G1 test   | G5 Flow         | G1/G2 screen QA | N/A                 |
| ZiG pay / EcoCash COD    | G1        | G5              | G1              | FX admin audit      |
| Agency disclosure + FDMS | G1        | G5 receipt link | G1              | Fiscal day worker   |
| Multi-vendor settle      | G2 ledger | G5              | G2              | Supplier statements |
| Slots + dispatch         | G3        | slot Flow       | G3              | delivery-android    |
| Age gate + audit         | G4        | G5 gate screen  | G4              | door check          |
| B2B hide informal        | G1 leak=0 | G5 filter       | G1              | —                   |
| AI no money              | G6        | N/A             | N/A             | ops drafts G6       |

## Part 13 — Proposed decisions (G-log) — grill-accepted where noted

Cite as G-n in tickets. **Grill Waves 1–2 closed 2026-08-14** — see

[DIAL\\_Groceries\\_Liquor\\_Grill\\_Session.md](#). Promote to v4 D-log only if ecosystem-wide.

| ID         | Decision draft                                                                                                                                                | Status                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>G-1</b> | Grocery/liquor is third Shop vertical in same ERP (no monorepo fork)                                                                                          | Accept (plan)                |
| <b>G-2</b> | Apply D-57 USD browse / ZiG pay / EcoCash COD buttons unchanged                                                                                               | <b>Grill-accepted</b>        |
| <b>G-3</b> | Meili index <code>grocery_offers_v1</code> separate from Spare                                                                                                | Accept (plan)                |
| <b>G-4</b> | Absorb domain into <code>@dial/catalogue</code> modules first; extract <code>packages/grocery</code> only if needed                                           | Accept (plan)                |
| <b>G-5</b> | Delivery fee as <code>DIAL_FEE</code> (option A); spoilage supplier until POD; Dial-caused cold-chain → Dial                                                  | <b>Grill-accepted</b>        |
| <b>G-6</b> | Escrow if multi-vendor $OR \geq T$ ; direct below $T$ ; liquor-high $\geq T_{liq}$ . <b>Provisional</b><br>$T=5000$ , $T_{liq}=10000$ USD minor (ops-tunable) | <b>Grill-accepted</b>        |
| <b>G-7</b> | Liquor formal-only suppliers at launch                                                                                                                        | <b>Grill-accepted</b>        |
| <b>G-8</b> | Food informal B2C: formal-first launch                                                                                                                        | <b>Grill-accepted</b>        |
| <b>G-9</b> | Packed fixed weights only in MVP; catch-weight deferred                                                                                                       | Accept (plan default stands) |

| ID          | Decision draft                                                                                                                                                                                | Status                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>G-10</b> | Extend <code>DeliveryDispatchWorkflow</code> (no parallel Grocery workflow SoR)                                                                                                               | Accept (plan)                              |
| <b>G-11</b> | WA grocery Flows <b>full MVP</b> (Part 8)                                                                                                                                                     | <b>Grill-accepted</b>                      |
| <b>G-12</b> | <b>No WHT</b> on grocery merchants; provisional payout <b>T+3</b> after capture/release                                                                                                       | <b>Grill-accepted</b>                      |
| <b>G-13</b> | Merchant holds liquor licence; platform terms + publish gates                                                                                                                                 | Provisional eng;<br><b>counsel pending</b> |
| <b>G-14</b> | Age default config 18 until counsel cites                                                                                                                                                     | Eng placeholder;<br>counsel for finals     |
| <b>G-15</b> | Open Food Facts optional enrichment off by default                                                                                                                                            | Accept (plan)                              |
| <b>G-16</b> | Delivery fee = deterministic multi-factor pricing engine ( <code>amountMinor</code> ); AI capability drafts rate-card/explanations only; no LLM live fees; no prod agent host ( <b>D-61</b> ) | <b>Grill-accepted</b>                      |

## Part 14 — Launch checklist & ops KPIs

### 14.1 Grocery/liquor launch checklist (additive to v4 Appendix C)

**Bold = blocks grocery customer-open for that SKU class.**

- Supplier contracts: agency, spoilage, oversell fee, indemnity, **liquor licence warranties**, food-safety certs for chilled/frozen
- **Counsel liquor opinion (Part 6.6) green — or liquor category dormant**
- **Age-gate + hours config live; audit trail live**
- **Licence publish gate live**
- Customer T&Cs: perishables, alcohol returns, 18-item disclosure audited on web/WA/native
- FDMS virtual device + `GOODS_*` / `DIAL_FEE` grocery paths tested
- Job Reserve / PspAdapter rails for grocery order namespace
- **IMTT never on checkout (pricing review signed)**
- Meili `grocery_offers_v1` + B2B informal leak=0
- Slot capacity + `DeliveryDispatch` activities
- delivery-android cold-chain + age-check UI
- WA Flows published + templates approved
- POTRAZ / D-32 still green for age data
- Insurance: product liability sized for foodborne claim **[Founder-input]**
- Threat Dragon grocery money + age-gate models; Semgrep CI green
- Tracer matrices G1–G5 green; G6 if AI exposed
- Living root docs updated in landing PR

## 14.2 Ops KPIs (dashboard from week one)

| KPI                      | Target posture                                   |
|--------------------------|--------------------------------------------------|
| Contribution / order     | <b>[Founder-input]</b> > 0 after stress spoilage |
| Ops minutes / order      | Trend down                                       |
| Fill rate                | ≥ Spare analog                                   |
| Confirm-fail rate        | Low; fee enforced                                |
| On-time slot %           | High                                             |
| Age-check fail %         | Monitor; refund SLA                              |
| Spoilage refund bps      | Within provision                                 |
| Repeat 90-day            | LTV:CAC track                                    |
| Advance-pay vs COD share | Float vs equity funding                          |

## Part 15 — Founder grill (answers-or-defaults)

Per `dial-grill-locks` : facts looked up in-repo; below are **human decisions**. Defaults in italics are **recommendations**, not locks.

**Living session:** `DIAL_Groceries_Liquor_Grill_Session.md` — **Waves 1–2 CLOSED** (2026-08-14). G1 thin vertical unblocked except liquor counsel.

| #   | Question                                | Recommended default                                                          | Status                                  |
|-----|-----------------------------------------|------------------------------------------------------------------------------|-----------------------------------------|
| Q1  | Launch geography / bands?               | <i>Harare metro single band first; reuse deliveryBandId</i>                  | <b>Accepted</b>                         |
| Q2  | FX display follows D-57?                | <i>Yes — USD browse/cart; ZiG at pay</i>                                     | <b>Accepted</b> → G-2                   |
| Q3  | Liquor licensing holder?                | <i>Merchant holds; DIAL terms + publish gates</i>                            | <b>Provisional eng; counsel pending</b> |
| Q4  | Age & hours numbers/ID types?           | <i>Config + min_age=18 pending counsel</i>                                   | <b>Accepted</b> eng; counsel for finals |
| Q5  | Informal grocery/liquor visibility?     | <i>Food formal-first; liquor formal-only</i>                                 | <b>Accepted</b> → G-7/G-8               |
| Q6  | Fee / delivery owner / spoilage bearer? | <i>DIAL_FEE + pricing engine; supplier until POD</i>                         | <b>Accepted</b> → G-5 / G-16            |
| Q7  | Merchant payout WHT / T+N?              | <i>No WHT; provisional T+3</i>                                               | <b>Accepted</b> → G-12                  |
| Q8  | Escrow vs direct?                       | <i>Direct below T; escrow multi-vendor/high ticket</i>                       | <b>Accepted</b> → G-6                   |
| Q9  | SLA / cold-chain / thaw refunds?        | <i>Same-day slots MVP; best-effort cold-chain + refund SOP before market</i> | Open (Wave 3 / ops)                     |
| Q10 | Supplier KYC / food-safety certs?       | <i>KYC before first payout; cert required chilled/frozen</i>                 | Open (Wave 3 / ops)                     |
| Q11 | WA in MVP?                              | <i>Full FLOW_GROCERY_ (Part 8)*</i>                                          | <b>Accepted</b> → G-11                  |



| #   | Question                                 | Recommended default                                       | Status                            |
|-----|------------------------------------------|-----------------------------------------------------------|-----------------------------------|
| Q12 | Brand: separate vs Shop tab?             | <i>Shop tab under Dial + branch branding; one account</i> | Open (Wave 3)                     |
| Q13 | Catch-weight?                            | <i>Packed fixed weights only MVP</i>                      | Plan default stands → G-9         |
| Q14 | B2B grocery MVP?                         | <i>B2C first; B2B formal-only if launched</i>             | Open (Wave 3)                     |
| Q15 | Multi-stop vs split jobs default?        | <i>One multi-stop job when same band/slot</i>             | Open (Wave 3)                     |
| Q16 | Take-rate launch bps / ladder?           | <i>[Founder-input] publish ladder to supply</i>           | Open (commercial)                 |
| Q17 | Escrow threshold T (USD minor)?          | <i>Provisional T=5000; T_liq=10000</i>                    | <b>Accepted provisional</b> → G-6 |
| Q18 | Liquor outside hours: browse ok or hide? | <i>Browse ok; ATC/checkout fail closed</i>                | Aligned with Q4                   |
| Q19 | ID image retention at door?              | <i>Boolean+method only unless counsel requires</i>        | Open (counsel/D-32)               |
| Q20 | Open Food Facts on by default?           | <i>Off; optional enrichment</i>                           | Accept → G-15                     |

**?** Format for live grill rounds — ask frontier only; refuse lock reopeners (C-5, Baileys, Mapbox SoR, AI money, DIAL\_OWNED, etc.).

## Part 16 — Architecture integration (seamless ERP)

### 16.1 Domain map

| Grocery concept | Existing SoR                                    |
|-----------------|-------------------------------------------------|
| Offer / product | Postgres catalogue + Meili                      |
| Cart            | Extend @dial/catalogue cart (vertical metadata) |
| Money           | @dial/payments + ledger + PspAdapter            |
| FDMS            | @dial/tax + adapter-fdms                        |
| Delivery        | @dial/delivery + Temporal                       |
| Promo           | @dial/promotions                                |
| WA              | @dial/adapter-whatsapp                          |
| Maps            | @dial/adapter-maps                              |

### 16.2 Mermaid — request path

Mermaid diagram (source — not rendered in PDF)

```
flowchart TB
  subgraph surfaces [Customer surfaces]
    WEB[gateway-web /grocery]
    NAT[Native Android+iOS]
    WA[WhatsApp Cloud API Flows]
  end
  subgraph edge [Edge]
    GW[gateway-web API + webhooks]
  end
  subgraph domain [DIAL domain packages]
    CAT["@dial/catalogue"]
    PAY["@dial/payments"]
    LED["@dial/ledger"]
    TAX["@dial/tax"]
    DEL["@dial/delivery"]
    PROMO["@dial/promotions"]
    AI["@dial/ai drafts"]
    ID["@dial/identity"]
  end
  subgraph adapters [Adapters]
    PSP["adapter-psp"]
    FDMS["adapter-fdms"]
    MAPS["adapter-maps"]
    WAA["adapter-whatsapp"]
  end
  WEB --> GW
  NAT --> GW
  WA --> WAA --> GW
  GW --> ID
  GW --> CAT
  GW --> PAY
  GW --> DEL
  CAT --> MEILI[Meilisearch grocery_offers_v1]
  PAY --> PSP
  PAY --> LED
  LED --> TAX
  TAX --> FDMS
  DEL --> TEMP[Temporal DeliveryDispatchWorkflow]
  DEL --> MAPS
```

16.3 Package boundary

**Absorb into @dial/catalogue + existing packages (recommended).** Reject parallel Medusa/Saleor. Optional later packages/grocery only if slotting/substitution logic proves catalogue-bloating.

16.4 Feature flags

DIAL\_FEATURE\_GROCERY=1 / PostHog; vertical tags; **not** DIAL\_BUSINESS\_BRANCH process fork that disables Spare/Tech.

Part 17 — Risks & non-goals

| Risk                                 | Mitigation                               |
|--------------------------------------|------------------------------------------|
| Agency threatened by owned inventory | Ban DIAL_OWNED; disclosure tests         |
| Food safety liability                | Contracts; SOP; insurance; counsel       |
| Liquor regulatory surprise           | Config gates; counsel before liquor SKUs |

| Risk                   | Mitigation                         |
|------------------------|------------------------------------|
| Dual SoR creep         | Stitch doctrine; money-path skill  |
| Settlement bugs        | Integer money; idempotent webhooks |
| Informal→B2B leak      | Shared filters + leak=0            |
| AI price hallucination | Zod; Promptfoo; capability review  |
| Slot overload          | Capacity + degrade messaging       |
| IMTT on checkout       | D-60 checklist                     |

**Non-goals:** principal grocer; renumber trains; Expo/Baileys/Mapbox SoR; Formance/Medusa ledger; Simulated auto-pay; promo cash-out; physical fiscal printers; parallel monorepo fork.

## Appendix A — Authority citations

| Lock                | Relevance                           |
|---------------------|-------------------------------------|
| C-4 / C-5           | Job Reserve; native not Expo        |
| D-38                | UX donors only                      |
| D-40 / D-40a / D-41 | WA; virtual FDMS; MVP Flows         |
| D-42                | promotions SoR                      |
| D-43                | PspAdapter                          |
| D-44 / D-45 / D-45a | maps + delivery SoR                 |
| D-47 / D-48         | IDOR + AppSec                       |
| D-49                | B2B hide informal                   |
| D-50                | Tech WHT ≠ auto grocery             |
| D-52 / D-56         | Tracer + grill                      |
| D-53 / D-54         | Factory; MetricContract             |
| D-57                | USD browse; ZiG pay; EcoCash COD    |
| D-58 / D-59 / D-60  | Agency; FDMS; IMTT; COD USD         |
| D-61                | AI capabilities; no prod agent host |

## Appendix B — Repo anchors

- `packages/catalogue`, `packages/payments`, `packages/delivery`, `packages/tax`, `adapters/fdms`, `adapters/whatsapp`
- `apps/gateway-web/src/app/spare/*` (pattern for `/grocery/*`)
- `docs/integrations/README.md`
- `docs/planning/DIAL_Tracer_DoD_Completion_Matrices.md`
- Pack §8 `spare_offers_v1` → mirror for `grocery_offers_v1`

Appendix C — Supplier grocery import columns (draft)

| Col | Field               | Required  | Notes                     |
|-----|---------------------|-----------|---------------------------|
| A   | Supplier SKU        | Yes       |                           |
| B   | Title               | Yes       |                           |
| C   | Barcode             | No        | OFF enrichment key        |
| D   | Category path       | Yes       | slash or pipe             |
| E   | Unit label          | Yes       | each / 500g / 1L          |
| F   | Packed weight grams | If weight | MVP packed                |
| G   | Net cost minor      | Yes       | integer                   |
| H   | Currency            | Yes       | USD default               |
| I   | Cold chain          | Yes       | ambient/chilled/frozen    |
| J   | Age gate            | Yes       | Y/N                       |
| K   | Stock valid until   | No        | default TTL by cold chain |
| L   | Liquor licence ref  | If liquor | publish gate              |

End of peer-depth plan. Next: founder answers Part 15 → counsel Part 6.6 for liquor → Dev Manager opens G1 thin-vertical ticket with DoD matrix → Build ( *diaL-tracer-slice* ). Living docs note only when Build lands (or when this plan is committed per Dev Manager).